Graphing systems of equations

1

a Infinitely many solutions

b One solution

c No solutions

d One solution

e No solutions

2

a $(3, -6)$

b $(6, -3)$

c $(-4, -2)$

3

a

x	-3	-2	-1	0	1	2	3
y	-15	-11	-7	-3	1	5	9

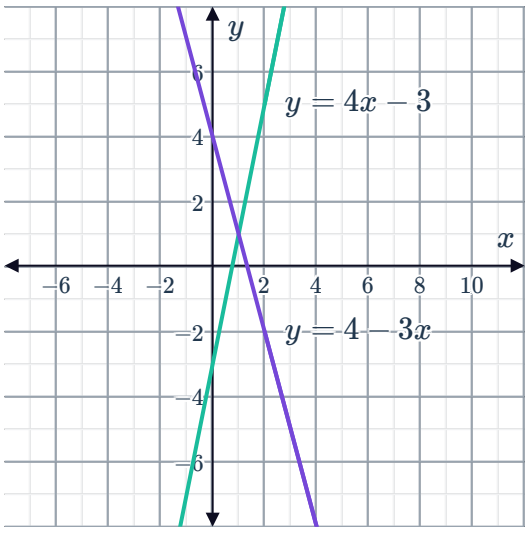
b

x	-3	-2	-1	0	1	2	3
y	13	10	7	4	1	-2	-5

c Slope $= 4$

d Slope $= -3$

e



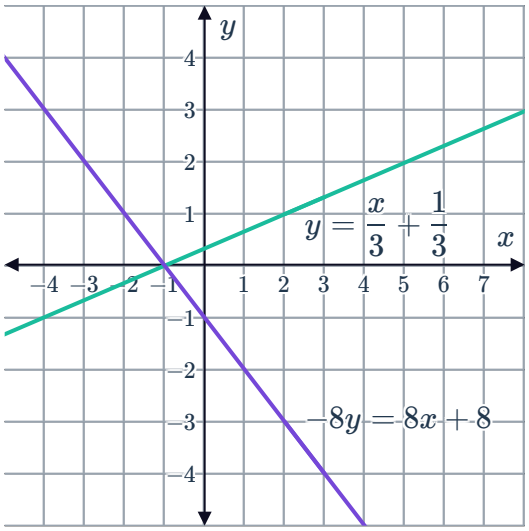
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4

a x-intercept: $(-1, 0)$
y-intercept: $\left(0, \frac{1}{3}\right)$

b x-intercept: $(-1, 0)$
y-intercept: $(0, -1)$

c



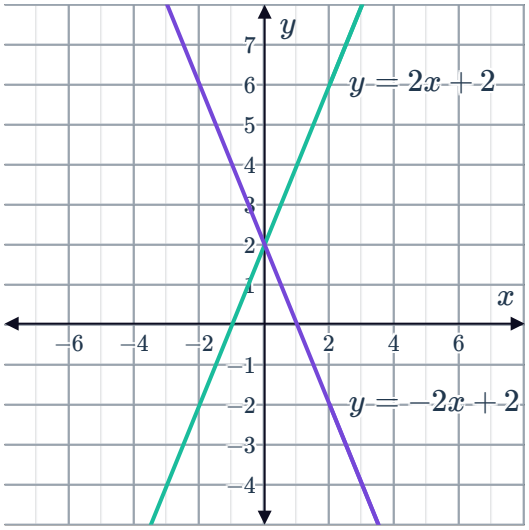
d $x = -1, y = 0$

5

a Slope: 2
y-intercept: $(0, 2)$

b x-intercept: $(1, 0)$
y-intercept: $(0, 2)$

c

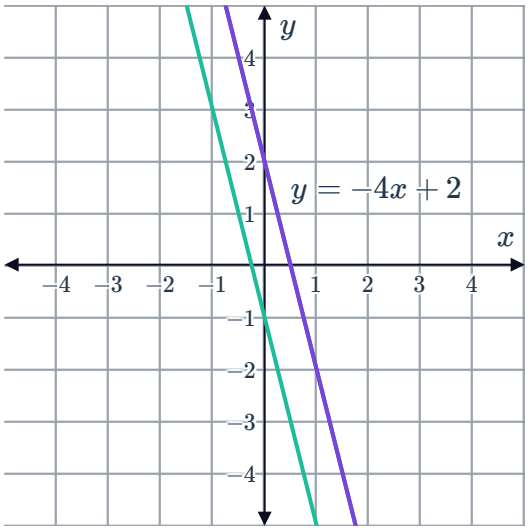


d $x = 0, y = 2$

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6 **a**

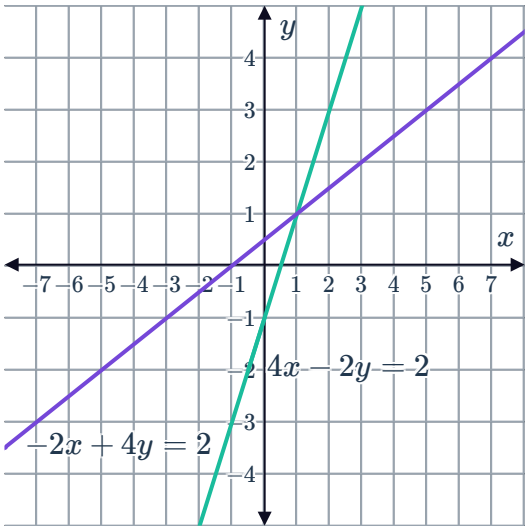
i



ii No

b

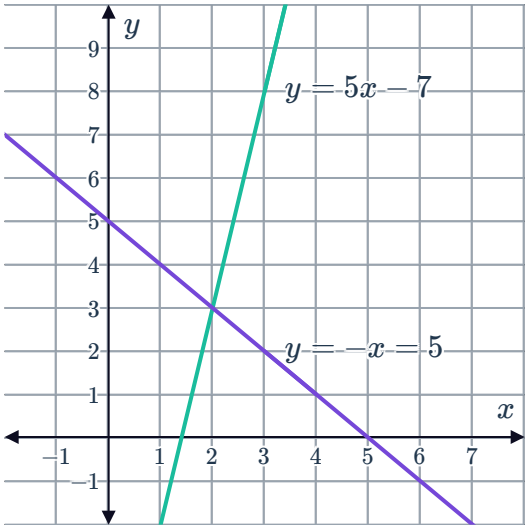
i



ii $x = 1, y = 1$

c

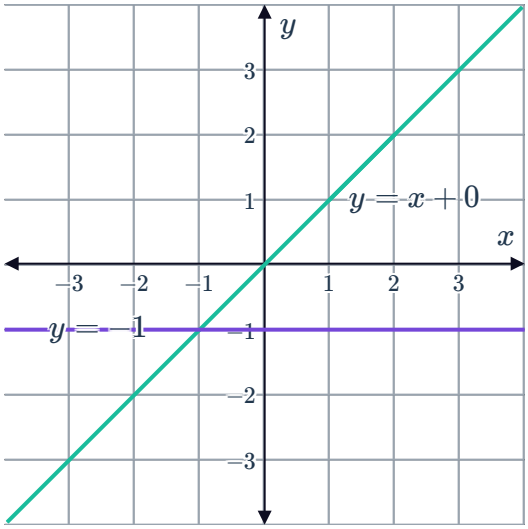
i



ii $x = 2, y = 3$

d

i



ii $x = -1, y = -1$

7 $y = -4x - 4$

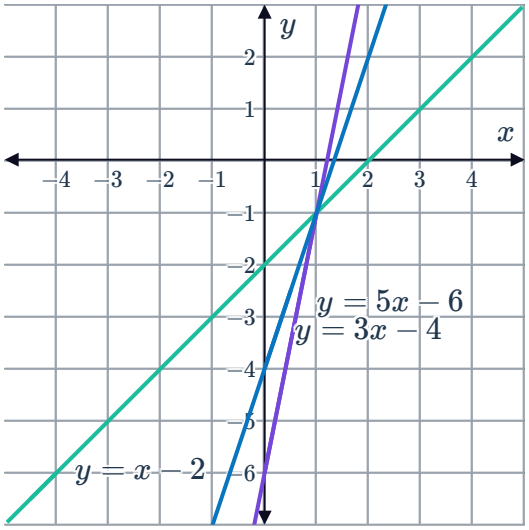
8 $b = 3$

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9

a $y = 3x - 4$

b

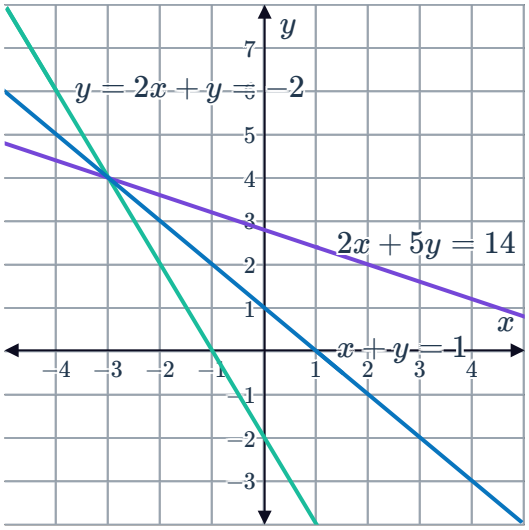


c $x = 1, y = -1$

10

a $x + y = 1$

b



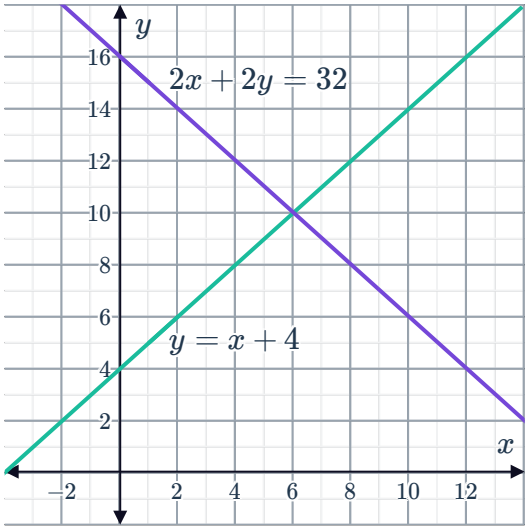
c $x = -3, y = 4$

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11

a $y = x + 4$
$2x + 2y = 32$

b



c Length $= 10$ ft, Width $= 6$ ft.

Additional questions

12 Answers will vary. For example:
Maria has 15 lbs of rice and cooks 4 lbs of rice every 3 days. Arjun has 12 lbs of rice and cooks 1 lbs of rice every 2 days.
The solution to the system $(14, 5)$ means that after 14 days, Maria and Arjun have the same amount of rice remaining.

13 The point of intersection on the graph has negative x and y values. In terms of the height of a plant over time, only positive values make sense. Since the two lines do not intersect in the first quadrant, they will never reach the same height on the same day.

14 If you need an exact solution to the system and the solution contains irrational numbers or fractions not easily readable on the scale of your graph, then solving the system by graphing is not feasible.

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